



Mehedi Hasan

AI/ML Engineer

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PROFESSIONAL EXPERIENCE

Senior Software Engineer, Technonext

Jun 2025 – Present

- ▷ Working on **Ticket Parsing** for airline/passenger documents, extracting structured fields such as **Passenger Data**, **Fare Data**, **Itinerary Details**, **Fare & Baggage Rules**, and **Fare Policy**.
- ▷ Building a **local LLM-powered parsing pipeline** using **vLLM** for high-throughput inference and low-latency serving.
- ▷ Designed a **heuristic validation layer** (rule/consistency checks) to verify LLM outputs, reduce hallucinations, and ensure schema-correct, business-safe extraction.
- ▷ Developing prompts, post-processing, and evaluation workflows for rapid iteration across new ticket formats and airlines. [vLLM](#) [Local LLM](#) [RAG](#) [Heuristic Validation](#) [Python](#) [FastAPI](#)

Senior AI Engineer, Next Solution Lab

Jun 2023 – Jun 2025

- ▷ Led **English DeepICR** (document understanding for contracts/invoices): built, trained, and tested **Text Detection**, **Layout Detection**, **Text Recognition**, and **Data-extraction**. Complex noise filtering (checkmarks/underlines), handwriting handling; document spell-correction via fuzzy + dictionary (Levenshtein) tuned to client lexicons. [Mask R-CNN](#) [OpenCV](#) [YOLOv7](#) [DBNet](#) [RCNN+Attention](#) [LCNN](#) [LayoutLM](#) [BROS](#)
- ▷ Optimized training and inference pipelines via model fine-tuning and parallel processing to improve throughput and latency. [PyTorch](#) [DDP](#) [CUDA/cuDNN](#) [joblib](#)
- ▷ Enabled **GPU multi-training** and **PDF batching** for large documents. [Keras](#) [PyTorch](#) [TensorFlow](#)
- ▷ Standardized evaluation with **automated reports** for PM/business sign-off. [XlsxWriter](#) [pandas](#)
- ▷ Packaged services with **Docker**, integrated **modular monolith** server-side APIs with client systems, and managed deployment/version control via **AWS CodeCommit**, **S3**, **EC2**, **ECR**, **Elasticsearch**, and **CloudWatch**. [Node.js](#) [React](#) [Material-UI](#) [AWS SDK](#)
- ▷ Monitored Japanese DeepICR platform; supported model updates and maintained routine regression tests.
- ▷ Implemented LoRA/PEFT-based **LLM** fine-tuning workflows and optimized inference acceleration pipelines for efficient model adaptation and deployment. [HuggingFace](#) [Transformers/TRL](#) [BitsAndBytes](#) [ONNX](#)
- ▷ Built a **RAG** search system for Japanese legal documents with memory and inference caching, enabling low-latency conversational Q&A. [LangChain](#) [ChromaDB](#) [Elasticsearch](#) [FastAPI](#)

AI Engineer, Next Solution Lab

Jun 2022 – Jun 2023

- ▷ Built the **Japanese Text-recognition** training pipeline with augmentation/evaluation and ~5,000-character coverage; created an **automated data-engineering & reporting** flow; maintained for new-data sprints. [RCNN+CTC](#)
- ▷ **English DeepICR**: trained/tested models for new datasets every sprint; standardized onboarding & regression tests.
- ▷ **DocQA (pre-LLM)**: Q&A for contracts/invoices documents; [BERT/RoBERTa](#) [Flask](#) [Streamlit](#)
- ▷ **Multilingual OCR**: Document OCR for (Arabic, Vietnamese, Thai, Indonesian). [Google Vision API](#)

- ▷ Deepfake detection exploration; STT/TTS evaluations; speaker verification. [FastAPI](#) [NVIDIA NeMo](#)

Associate AI Engineer, Next Solution Lab

Jun 2021 – Jun 2022

- ▷ **Key-Value Extraction Module:** deterministic OpenCV rule engine extracting **128 fields** on **44** document clusters; authored **SRS**, exposed **API**, and defined/tested plans; owned **text-line** & **text-box update** integration for readability. [OpenCV](#) [REST API](#)
- ▷ **Smart Farming (POC):** camera-only cattle monitoring under a no-RFID policy; delivered fast-deploy models and proposal. [Detectron2](#) [SORT](#)
- ▷ **ID/OCR:** driving licence extraction. [YOLOv5](#) [FastAPI](#) [Streamlit](#)

TECHNICAL SKILLS

Programming	Python, JavaScript, Node.js, C#, Java
Frameworks & Libraries	PyTorch, TensorFlow, Keras, Transformers, FastAPI, Streamlit, scikit-learn, OpenCV, LangChain, Spacy
ML & Data Tools	pandas, NumPy, joblib, XlsxWriter, matplotlib, seaborn, Selenium, Playwright
Cloud & DevOps	AWS (EC2, ECS, S3, Lambda, CloudWatch, CodeCommit/ECR), Docker, CI/CD
Database & Search	MySQL, MSSQL, Elasticsearch, OpenSearch, ChromaDB
Frontend	React, Material-UI, React Router
GPU & Parallelization	CUDA, cuDNN, PyTorch DDP
Version Control	Git, Agile/Scrum

AI MODELS & ARCHITECTURES

Computer Vision	YOLOv7/YOLOv5, DBNet, Mask R-CNN, RCNN+CTC, LCNN
Document & Layout Understanding	LayoutLM, BROS, DocFormer
Language & LLMs	BERT, RoBERTa, Transformers, LoRA/PEFT, DPO

EDUCATION

Ahsanullah University of Science and Technology (AUST) <i>Bachelor of Science in Computer Science and Engineering — CGPA: 3.59</i>	<i>Dhaka, Bangladesh</i> <i>Apr. 2016 – Jan. 2021</i>
SOS Hermann Gmeiner School and College <i>Higher Secondary Certificate — GPA: 5.00</i>	<i>Dhaka, Bangladesh</i> <i>2013 – 2015</i>
Mirpur Bangla High School and College <i>Secondary School Certificate — GPA: 5.00</i>	<i>Dhaka, Bangladesh</i> <i>2003 – 2013</i>

PROJECTS

Shorol Notes — AI-Powered Note-Taking 🧠

- ▷ Voice notes with AI transcription/summarization; calendar sync; Bangla-first UX.
- ▷ Pluggable backends (OpenAI/Ollama/Hugging Face); responsive, offline-friendly.

Tech: [React](#), [TypeScript](#), [Vite](#), [Node.js/Express](#), [Tailwind CSS](#), [OpenAI API](#), [Hugging Face](#), [Ollama](#)

AI-Powered Research Agent (RAG + LLM) 🔗

- ▷ ReAct-style tool-using agent with PubMed, Wikipedia, ArXiv, and web search.
- ▷ FastAPI UI + terminal; configurable prompts/models for experiments.

Tech: [Python](#), [FastAPI](#), [LangChain](#), [Transformers](#), [Hugging Face](#), [SerpAPI](#)

Japanese Lawyer Assistant 🔗

- ▷ Legal Q&A over Japanese corpus using RAG (retriever + FAISS) with instruct LLM.
- ▷ Knowledge-base pipeline; concise answers with custom prompts.

Tech: [Python](#), [LangChain](#), [FAISS](#), [FastAPI](#), [ELYZA LLaMA-2](#), [Hugging Face](#)

RESEARCH EXPERIENCE

Bengali Intent Classification with Generative Adversarial BERT 🔗

First Author, IEEE Xplore, 2023

Developed **GAN-BnBERT**, a generative adversarial [BERT](#)-based model for Bengali intent classification. Introduced the **BNIntent30** dataset with 30 intent classes, demonstrating superior performance over [Bi-LSTM](#) and baseline [BERT](#) models.

Design of an Arrhythmia Classification Algorithm Using 2-D Convolutional Neural Network 🔗

First Author, Undergraduate Thesis, AUST, 2021

Proposed a [CNN](#)-based arrhythmia detection algorithm from [ECG](#) signals, comparing performance with [VGGNet](#) for improved beat classification accuracy.

ACADEMIC PROJECTS

Hospital Management 🔗 | Web Application

Developed a full-stack hospital management system using [ASP.NET MVC](#), [C#](#), [Razor](#).

Tolet-Dhaka 🔗 | Web Application

Built a flat-renting website with [HTML5](#), [CSS](#), [PHP](#); enhanced interactivity with [JavaScript](#).

Bank Management System 🔗 | Database Application

Implemented transactional features with [Java](#), [Java GUI](#), [MySQL](#).

Dhaka City Bus Routes 🔗 | Android Application

Created a route-finder app using [Java](#), [Google Maps API](#), [JSON](#).

Railway Management System 🔗 | Database Application

Developed client-server components in [PL/SQL](#).

NON-TECHNICAL SKILLS

- ▷ Stakeholder communication (PM/business)
- ▷ Agile/Scrum
- ▷ Work Breakdown Structure (WBS)
- ▷ SRS & documentation
- ▷ Versioned runbooks
- ▷ Technical presentations

LANGUAGE SKILLS

Bangla (Native); English (Professional Working Proficiency).